

Seminar Quiz 3

Empirical Analysis for Policymakers

Sciences Po Paris - Reims

30 minutes. No extra papers. You are free to use calculators.

Multiple Choice Questions (MCQs)

1. Which of the following is an assumption of the Difference-in-Differences (DiD) estimator?
 - A) The treatment and control groups have identical trends in outcomes in the absence of treatment.
 - B) The treatment group always has a higher outcome than the control group.
 - C) The treatment effect is heterogeneous across individuals.
 - D) The treatment and control groups have identical levels of outcomes before the treatment.
2. In Regression Discontinuity (RD), which of the following is true about the assignment variable?
 - A) It must be binary.
 - B) It must have a clear cutoff value.
 - C) It must be normally distributed.
 - D) It must be uncorrelated with the outcome variable.
3. The graphical method for RD estimation involves:
 - A) Plotting the residuals.
 - B) Plotting the outcome variable against the assignment variable.
 - C) Plotting the treatment effect.
 - D) Plotting the independent variable against the dependent variable.
4. In the context of DiD, if the assumption of parallel trends is violated, what is the consequence?
 - A) The DiD estimator will still be unbiased.
 - B) The DiD estimator will be biased.
 - C) The RD estimator will be biased.
 - D) There is no effect on the estimator.
5. What is the main purpose of using a Regression Discontinuity design?
 - A) To establish external validity.
 - B) To identify causal effects using a discontinuity in the assignment variable.

- C) To test for omitted variable bias.
- D) To correct for measurement error.

True/False Questions

1. In a Difference-in-Differences analysis, the key identifying assumption is that the treatment and control groups would have followed parallel trends in the absence of treatment.
2. In Regression Discontinuity, treatment assignment is determined randomly.
3. The DiD estimator uses both before-and-after and treatment-vs-control comparisons to identify the treatment effect.
4. In RD designs, the treatment effect is estimated as the difference in outcomes at the cutoff point.
5. Both DiD and RD are used to identify causal effects in non-experimental settings.

Short Essay Question

You are evaluating the impact of a government-subsidized childcare program on female labor force participation using a Difference-in-Differences approach. Discuss the steps you would take to implement this model and explain the main assumption required for the identification of the treatment effect. Briefly outline how you would validate this assumption. (2 points)

Mathematical Questions

1. **Difference-in-Differences Estimation:** A city implements a policy to reduce traffic congestion by introducing a congestion charge in the downtown area. You are tasked with evaluating the policy's impact on traffic volume using a Difference-in-Differences approach.

Below is the average daily traffic volume (in thousands of vehicles) in the downtown area (Treatment) and a neighboring city (Control) before (t_1) and after (t_2) the policy:

Area / Time	t_1 (Y)	t_2 (Y)
Downtown (Treatment)	80	60
Neighboring City (Control)	90	85

- a) Calculate the DiD estimator for the policy's effect on traffic volume. (1 point)
- b) Interpret the result. (1 point)
- c) Write down the equation for the DiD regression and show the DiD coefficient. (1 point)
- d) Calculate the time trend $\lambda_{t_2} - \lambda_{t_1}$. (1 point)
- e) Interpret the time trend and discuss the main assumption of the DiD approach that is related with it. (1 point)

2. **Regression Discontinuity Estimation:** Suppose we are studying the effect of a scholarship program on academic performance. Students with test scores above 70 are eligible for the scholarship. The following regression model is estimated:

$$Y_i = \alpha + \tau D_i + \beta(\text{Score}_i - 70) + \epsilon_i$$

where Y_i is the outcome (academic performance), D_i is a binary variable indicating treatment (1 if score ≥ 70 , 0 otherwise).

- a) Explain how τ captures the treatment effect. (1 point)
- b) Plot a representative graph of the data and show a visual confirmation of the RD assumption. (2 points)